Yang Wu

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EDUCATION

2023-2027 B.S. in Mathematics | Soochow University
Specialization: Dynamical System and Mathematical Modeling
Taipei City, Taiwan

2023-2027 B.A. in Music Composition | Soochow University
Taipei City, Taiwan

2024 Neuropsychiatric Disorder Summer Associate School | National Taiwan University
held by International Brain Research Organization (IBRO)
Taipei City, Taiwan

2024 Cognitive Neuroscience Summer School | National Central University
Taoyuan City, Taiwan

2022-2023 B.A. in Music (Piano Performance) | Florida State University
Tallahassee, FL, United States of America

RESEARCH

2026/2 ~Present **Sam Gershman lab & Kanaka Rajan lab | Harvard University**
Research Assistant, Department of Psychology / Harvard Medical School / Kempner Institute for the study of Natural and Artificial Intelligence
Advisor: John Vastola

- **Characteristics of Multi-Timescale Efficient-Coding Neural Network under Energy Efficiency Constraint**: Designed and ran CPC/InfoNCE experiments confirming three theoretical predictions about how representation timescales emerge in efficient-coding networks with temporal smoothness energy costs
- **Computational Model of Molecular Memory** *[Gershman lab]*: Developing computational model of memory storage at the molecular level, inspired by “The molecular memory code and synaptic plasticity” (Gershman et al., 2023); Building neural-symbolic architecture using Combinatory Logic primitives within neurons, inspired by “An RNA-Based Theory of Natural Universal Computation” (Akhlaghpour et al., 2022)

2025/7 ~Present **James Whittington lab | University of Oxford**
Research Assistant, Department of Experimental Psychology
Advisors: James Whittington and Joseph Warren

- **Identified specific neural algorithms the RNN uses in sequential memory task**: Extending previous work “A tale of two algorithms” (Whittington et al., 2024) to model how the hippocampus reorganizes its cognitive map during sequence learning
- **Mechanistic interpretability of RNN representations**: using linear decoders to test competing hypotheses about observation encoding. Identified the "stretching" of neural subspaces during sequence learning of RNN that generalized across multiple loop lengths, and quantified different learning phases’ decoder alignment via cosine similarity

2025/7 ~Present **Marcelo Mattar lab | New York University**
Research Assistant, Department of Psychology
Advisor: Marcelo Mattar

- **Modeling Episodic Simulation through Memories Recombination**: Built key-value episodic memory module integrated with GRU controller to model human episodic simulation, i.e. how humans recombine episodic memories from the past to implement future planning or imagination (Schacter et al., 2007)
- Designed three phases reinforcement learning task (exploration → mental simulation → execution) to study how episodic memories from past experiences support future planning and imagination
- Established that Phase2 mental simulation improves Phase3 execution; investigating whether this benefit reflects explicit sequential plan formation during mental simulation or mere enrichment of key-value memory representations

2025/7 ~2025/12 **Kanaka Rajan lab | Harvard University**
Research Assistant, Harvard Medical School / Kempner Institute for the Study of Natural and Artificial Intelligence
Advisors: Ryan Badman and John Vastola

- **Interpreting Planning-Liked Behavior in Naturalistic Foraging RL Agents**: Analyzing emerged planning-like behaviors ethologically in model-free reinforcement learning agents within naturalistic environments
- Implemented Sparse Autoencoder, Generalized Linear Model, Decoder and Auto-Correlation to probe into agents internal representations and external behavior
- Publication: Contributed code and analysis to **Deep RL Needs Deep Behavior Analysis: Exploring Implicit Planning by Model-Free Agents in Open-Ended Environments.** ***Advances in Neural Information Processing Systems*****, 2025**; formally recognized in acknowledgements

2024/7 ~2025/7 **Yu-Wei Wu lab | Academia Sinica**
Research Assistant, Institute of Molecular Biology
Advisor: Yu-Wei Wu, Yiko Chen

- **Decomposing Inter-Regional Neural Dynamics during Mouse Motor Control**: Using Current-Based Decomposition Method (CURBD) and Dissociative Prioritized Analysis of Dynamics (DPAD) to model and describe the inter-regional communication between M1, M2 and S1 during mouse on-cue reaching task

2024/2 ~2025/3 **Yu-Te Wang lab | Academia Sinica**
Research Assistant, Research Center for Information Technology Innovation
Advisor: Yu-Te Wang

- **Development of 3D Printed Semi-Dry EEG Electrodes**: Building EEG electrode with 3D Printing. Human EMG, EEG Data Collection, CompactCNN for EEG Data analysis. Leading to multiple publications in conference and IEEE journal
- VR/AR-Based HCI Telerehabilitation Systems for EMG Motor Recovery using Unity

PUBLICATIONS

R. Simmons-Edler, R. P. Badman, F. B. Berg, R. Chua, J. J. Vastola, J. Lunger, W. Qian, K. Rajan "Deep RL Needs Deep Behavior Analysis: Exploring Implicit Planning by Model-Free Agents in Open-Ended Environments" *Advances in Neural Information Processing Systems*, 2025.; contribution formally recognized in acknowledgements.

C.-M. Chung, Y.-T. Wang*, J.-B. Lu, Y.-C. Hsu, Y.-J. Su, **Y. Wu**, C.-F. Hung "3D printed watermill-like semi-dry electrodes for BCI applications", *IEEE Transaction on Neural Systems and Rehabilitation Engineering*, 2025.

Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-T. Wang*, T.-H. Cheong "HCI Technologies in rehabilitation: enhancing physical therapy with sEMG, VR and LLM-Generated music", *Society for Neuroscience Annual Meeting*, 2024.

Y. Wu*, Y.-J. Su, J.-B. Lu, Y.-H. Chen, Y.-T. Wang* "RehabVerse: Mixed Reality Solution for Telerehabilitation", *National Biotechnology Research Park Pitch Day*, 2024.

Y.-L. Chu, C.-C. Tsao, C.-M. Chung, C.-F. Hung, Y.-Y. Lee, J.-B. Lu, **Y. Wu**, Y.-H. Chen, Y.-T. Wang* "BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification", *IEEE Brain Discovery & Neurotechnology Workshop*, 2024.

AWARDS / GRANTS / EXTRACURRICULAR

2026	Ministry of Education Study Aboard programme for Future Scholars in Humanities and Social Sciences (US$50,000)
2025	**Computational and Systems Neuroscience Conference (COSYNE) Undergraduate Travel Grants** (US$1,500)
2025	Chu-Chin Hsu Scholar
2025	Dean's list (Academic Excellence Award) * 2
2024	Distinction Award, International Information and Communication Technology Innovation Services Award (BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification)
2024	National Science and Technology Council Neuroscience Camp
2024	Dean's list (Academic Excellence Award) * 1
2024	Academia Sinica Statistical Science Camp
2024	Finalist, National Biomedical Engineering Award (BrainPrint: Innovative Head-Mounted EEG Technology for Secure Personal Identification)

COURSES / SKILLS

Important Courses Taken (Grades are based on the scale of 100): **Computational Neuroscience (Grad Level) (95/100), Statistical Mechanics (Grad Level) (87/100)**, Advanced Calculus (99/100), Linear Algebra (96/100), Programming in Life Science (97/100), Life Science (95/100)

Computation&Code: Python, PyTorch, Blender, 3D Printing, Git, Matlab

Modeling&Analysis: Neural Network, Transformer, Reinforcement Learning, PPO, Sklearn Decoder, Model Optimization, Sparse Autoencoder, Generalized Linear Model, EEG/EMG Data Collection

LANGUAGES

Mandarin: native
English: fluent/native
Japanese: beginner
French: beginner

MUSIC

Milonga – Latin-American piece for Cello and Piano https://vimeo.com/1036267350?share=copy

Zhang Beihai said good-bye to his father – Solo Piano https://vimeo.com/1036267291?share=copy

Weathering With You Movie Trailer – Re-scored by Me
https://vimeo.com/1036267308?share=copy